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Probabilistic Action Prediction of Human Drivers Using Dynamic Bayesian Networks

Master Thesis 1111

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List of Abbreviations

AV Autonomous Vehicle

HDV Human-Driven Vehicle

DBN Dynamic Bayesian Network

DAG Directed Acyclic Graph

MAP Maximum A Posteriori

AI Artificial Intelligence

THW Time Headway

TTC Time to Collision

KI Künstliche Intelligenz

CPU Central Processing Unit

SAV Semi-Autonomous Vehicle

V2X Vehicle-to-Everything

RV Random Variable

Glossary

Autonomous Vehicle (AV) A vehicle that can drive by itself without human control. It uses sensors and artificial intelligence to understand the surrounding environment and make driving decisions.

Human-Driven Vehicle (HDV) A vehicle that is controlled by a human driver who performs all driving tasks such as steering, accelerating, and braking.

Dynamic Bayesian Network (DBN) A probabilistic graphical model that uses a directed acyclic graph to represent time-series data, where each node represents a random variable and the edges show how these variables depend on each other within and across time steps.

Directed Acyclic Graph (DAG) A graph made up of nodes and directed edges between them, where the edges do not form any closed loops.

Mixed Traffic A traffic situation where both autonomous vehicles and human-driven vehicles operate together.

Trajectory The time-ordered sequence describing the movement of a vehicle over time.

Time Headway (THW) The time needed for a vehicle to reach the current position of the vehicle ahead if it maintains its current velocity.

Time to Collision (TTC) The time needed for a vehicle to collide with the vehicle ahead if both of them maintain their current velocity.

Latent State A variable that is not directly observed but is inferred from data.

Emission Model The probability distribution that describes how observations are generated from latent states.

Observation The data that is directly measured from the system, represented as feature vectors.

State Transition Probability of transitioning from one latent state to another.

Inference The process of determining the most likely latent states from the observed data.

Belief State The probability distribution over the latent states given the observed data at a specific time step.

Likelihood The probability of the observed data under a given model.

Prior The probability distribution over the latent states before observing any data.

Posterior The probability distribution over the latent states after observing the data.

Maximum A Posteriori (MAP) The estimate of the most likely latent state, given the observed data. It is the state that maximizes the posterior distribution.

Confusion Matrix A matrix showing the performance of a classification model, across the true and predicted classes.

Precision The ratio of true positive predictions to the total predicted positives. It indicates the accuracy of positive predictions.

Recall The ratio of true positive predictions to the total actual positives. It indicates the completeness of positive predictions.

F1 Score The harmonic mean of precision and recall.

Entropy A measure of uncertainty in a probability distribution. It also quantifies the amount of information needed to describe the distribution.

Random Variable (RV) A variable that can take on different values, each with an associated probability.

Kurzfassung

Künstliche Intelligenz (KI) verändert autonome Systeme durch bedeutende Fortschritte in den Bereichen Sensorik, Kommunikation und Rechenleistung. Obwohl vollständig autonome Fahrzeuge (Autonomous Vehicles (AVs)) voraussichtlich immer häufiger werden, werden sie auf absehbare Zeit gemeinsam mit von Menschen gesteuerten Fahrzeugen (Human-Driven Vehicles (HDVs)) am Straßenverkehr teilnehmen. Um in diesem gemischten Verkehrsumfeld sicher zu operieren, müssen autonome Systeme in der Lage sein, menschliche Fahrhandlungen zu modellieren und vorherzusagen.

Diese Arbeit adressiert dieses wichtige vorgelagerte Problem durch die Entwicklung eines transparenten, probabilistischen Rahmens zur Vorhersage hochrangiger menschlicher Fahrhandlungen unter Verwendung dynamischer Bayes'scher Netze (Dynamic Bayesian Networks (DBNs)). Ziel dieser Forschung ist die Entwicklung eines Modells, das zukünftige Fahreraktionen abschätzen kann und gleichzeitig Unsicherheiten in einer interpretierbaren Form darstellt. Ein solches Modell kann innerhalb eines AV-Systems als Belief-Modell dienen und Informationen für nachgelagerte Planungs- und sicherheitsrelevante Module bereitstellen.

Die Studie verwendet den naturalistischen Autobahnfahrdatensatz highD, um zeitliche Muster menschlichen Fahrverhaltens zu erlernen. Fahrzeugtrajektorien werden in sequenzielle, fensterbasierte Beobachtungen umgewandelt, die relevante Bewegungs- und Interaktionsinformationen über die Zeit erfassen. Diese Beobachtungen fassen zusammen, wie sich ein Fahrzeug und der umgebende Verkehr über kurze Zeitintervalle entwickeln. Anschließend wird ein DBN trainiert, um die zeitliche Entwicklung des Fahrerverhaltens zu modellieren und Aktionswahrscheinlichkeiten aus eingehenden Beobachtungen abzuleiten. Das vorgeschlagene DBN verwendet eine hierarchische latente Struktur, in der Aktionen von Fahrregimen abhängig sind. Genauer gesagt repräsentiert das Modell zwei latente Regime, ein Regime mit geringer Interaktion und ein Regime mit hoher Interaktion, und sagt Aktionszustände innerhalb jedes Regimes vorher. Dadurch kann das Modell sowohl das vorhergesagte Fahrmanöver als auch den Interaktionskontext darstellen, in dem es auftritt.

Eine zentrale Stärke dieses Ansatzes liegt in seiner transparenten probabilistischen White-Box-Struktur. Im Gegensatz zu vielen Black-Box-Lernmethoden erzeugt das DBN seine Vorhersagen über explizite probabilistische Beziehungen, die mithilfe der Wahrscheinlichkeitstheorie interpretiert werden können. Dies ist insbesondere für sicherheitsrelevante Anwendungen von Bedeutung, bei denen Interpretierbarkeit und die Darstellung von Unsicherheiten wichtig sind. Das Modell unterstützt daher nicht nur die Vorhersageleistung, sondern auch die Erklärbarkeit, was bei der Betrachtung solcher

Systeme für sicherheitskritische Verkehrsanwendungen von Vorteil ist. Die Studie leistet einen Beitrag zur Forschung im Bereich des autonomen Fahrens, indem sie Aktionsvorhersage, zeitliche probabilistische Schlussfolgerung und Interpretierbarkeit in einem einheitlichen Rahmen kombiniert.

Die Ergebnisse zeigen, dass das Modell in der Lage ist, aussagekräftige Verhaltensmuster aus naturalistischen Verkehrsdaten zu lernen und vielversprechende Manövervorhersagen über einen zukünftigen Zeithorizont zu generieren. In der aktuellen Evaluation erreichte das Modell eine exakte Next-Step-Genauigkeit von 46,4 %, eine Makropräzision von 44,3 % sowie eine Trefferquote von 68,0 % über einen Vorhersagehorizont von zwei Sekunden. Eine Vorhersage zum Zeitpunkt t wird dabei als Treffer gezählt, wenn das vorhergesagte Manöver innerhalb der nächsten fünf Fensterschritte auftritt, was etwa zwei Sekunden entspricht. Die durchschnittliche Time-to-Hit betrug 0,50 s und der Median 0,40 s, was zeigt, dass viele korrekte Vorhersagen früh innerhalb des Vorhersagehorizonts auftreten. Diese Ergebnisse deuten darauf hin, dass das Modell zukünftige menschliche Fahreraktionen innerhalb eines praktisch nutzbaren kurzfristigen Zeitintervalls antizipieren kann, anstatt nur einen einzelnen exakten zukünftigen Zeitpunkt vorherzusagen. Implementierungs-Benchmarks zeigen zudem geringe Online-Inferenzkosten mit einer mittleren CPU-Latenz von 0,164 ms pro Fahrzeug für Belief-Update und nächste Aktionsvorhersage.

Insgesamt zeigen die Ergebnisse, dass DBNs ein vielversprechender Ansatz für die probabilistische Vorhersage menschlicher Fahreraktionen sind. Die Studie demonstriert, dass hochrangige menschliche Fahrmanöver mithilfe naturalistischer Verkehrsdaten in einer interpretierbaren und zeitlich strukturierten Weise modelliert werden können. Dies stellt einen wichtigen Schritt hin zu einem besseren Verständnis menschlichen Fahrverhaltens im gemischten Verkehr dar und kann AV-Systeme dabei unterstützen, das Verhalten umliegender menschlicher Fahrer besser einzuschätzen.

Stichwörter: Vorhersage menschlichen Fahrverhaltens, Autonomes Fahren, Von Menschen gesteuerte Fahrzeuge, Dynamische Bayes'sche Netze, Manövervorhersage, Probabilistische Modellierung, Gemischte Verkehrsumgebungen

Abstract

Artificial Intelligence (AI) is transforming autonomous systems through major advances in sensing, communication, and computation. Although fully Autonomous Vehicles (AVs) are expected to become more common, they will share the road with Human-Driven Vehicles (HDVs) for the foreseeable future. To operate safely in this mixed-traffic setting, autonomous systems must be able to model and predict human driver actions.

This work addresses this important upstream problem by developing a transparent, probabilistic framework for predicting high-level human driving actions using Dynamic Bayesian Networks (DBNs). The purpose of this research is to develop a model that can estimate future driver actions while also representing uncertainty in an interpretable form. Such a model can act as a belief model inside an AV framework and provide information for downstream planning and safety-related modules.

The study uses the highD naturalistic highway driving dataset to learn temporal patterns of human driving behavior. Vehicle trajectory data are converted into sequential window-based observations that capture relevant motion and interaction information over time. These observations summarize how the vehicle and its surrounding traffic evolve over short time intervals. A DBN is then trained to model the temporal evolution of driver behavior and to infer action probabilities from incoming observations. The proposed DBN uses a hierarchical latent structure in which actions are conditioned on driving regimes. More specifically, the model represents two latent regimes, low-interaction and high-interaction, and predicts action states within each regime. This allows the framework to represent both the predicted maneuver and the interaction context in which it occurs.

A key strength of this approach is that it provides a transparent, white-box probabilistic structure. Unlike many black-box learning methods, the DBN makes predictions through explicit probabilistic relationships that can be interpreted using probability theory. This is valuable in safety-relevant applications, where interpretability and uncertainty representation are important. The model therefore supports not only prediction, but also explainability, which is useful when such systems are considered for use in safety-critical transport settings. The study contributes to science and engineering by combining action prediction, temporal probabilistic reasoning, and interpretability in a single framework for autonomous driving applications.

The results show that the model can learn meaningful behavioral patterns from naturalistic traffic data and generate promising maneuver predictions over a future horizon. In the current evaluation, the model achieved 46.4% exact next-step accuracy, 44.3% macro precision, and a 68.0% hit rate over a 2-second prediction horizon. Here, a prediction

made at time t is counted as a hit if the predicted maneuver appears at any point within the next 5 window steps, corresponding to about 2 seconds. The mean time-to-hit was 0.50 s, and the median time-to-hit was 0.40 s, showing that many correct predictions occur early within the horizon. These results suggest that the model can anticipate future human driver actions within a practically useful short-term interval rather than only at one exact future instant. Implementation benchmarks also indicate low online inference cost, with a mean single-vehicle Central Processing Unit (CPU) latency of 0.164 ms for belief update and next-action prediction.

Overall, the findings indicate that DBNs are a promising framework for probabilistic human driver action prediction. The study shows that high-level human driving maneuvers can be modeled in an interpretable and temporally structured way using naturalistic traffic data. This provides a useful step toward better behavior understanding in mixed traffic and may support AV systems that need to reason about nearby human drivers.

Keywords: Human Driver Behavior Prediction, Autonomous Driving, Human-Driven Vehicles, Dynamic Bayesian Networks, Maneuver Prediction, Probabilistic Modeling, Mixed Traffic Environments

1. Introduction

Automation is influencing almost every part of daily life, and transportation is one of the areas changing most rapidly. Autonomous Vehicles (AVs) are at the forefront of this change, revolutionizing the transportation sector. Their development has accelerated because of recent progress in Artificial Intelligence (AI), sensing systems, computing, and communication technologies. These innovations could enhance traffic efficiency, increase travel comfort, improve road safety, and, most importantly, reshape future mobility.

However, fully autonomous traffic systems do not yet exist anywhere in the world. AVs will have to operate in mixed traffic together with Semi-Autonomous Vehicles (SAVs) and Human-Driven Vehicles (HDVs). This creates important challenges because different road users provide the different level of information. AVs typically rely on rich on-board sensing and potentially have Vehicle-to-Everything (V2X) communication, while HDVs do not explicitly communicate their future intentions and often behave in a stochastic way [1][2]. As a result, the internal goals, decisions, and reactions of human drivers cannot be observed directly. They must instead be inferred from observable signals such as vehicle motion, lane changes, relative positions, and interactions with nearby vehicles.

For this reason, human driver behavior prediction is an important problem in autonomous driving. If an AV can estimate what nearby human drivers are likely to do next, it can make safer and better-informed decisions. For example, knowledge of likely braking, lane-changing, or car-following behavior can support downstream tasks such as risk assessment, trajectory prediction, and motion planning [3].

Over the last two decades, the researchers have developed three broad families of behavioral models for road users: rule-based models, probabilistic models, and learning-based methods [4] [5] [6]. In parallel, several game-theoretic decision-making paradigms for AVs have emerged [7]. Rule-based models are simple and easy to understand, but they are often too rigid to capture the variability of real human behavior. Learning-based methods can learn complex patterns from large datasets, but many of them are difficult to interpret and often behave like black boxes. Probabilistic models provide a useful middle ground because they can explicitly represent uncertainty, capture temporal dependencies, and still remain interpretable.

This thesis focuses on probabilistic modeling of high-level human driving behavior in highway traffic. Rather than building the full control or planning stack of an AV, this work addresses an important upstream problem: predicting the likely future maneuvers of surrounding human drivers. The goal is to develop a model that can estimate future driver actions while also representing uncertainty in a clear and interpretable way.

To achieve this, the thesis uses a Dynamic Bayesian Network (DBN) to model the temporal relationship between observable driving features and hidden behavioral states. Since driver intentions are not directly observable, the model represents them through latent variables and infers them probabilistically from trajectory-based observations [8]. In particular, the DBN is designed to capture latent factors such as driving context and action dynamics, and to produce probabilistic predictions of future high-level maneuvers. The model is trained and evaluated using the highD dataset, a naturalistic highway driving dataset containing detailed vehicle trajectory recordings [9]. From these trajectories, relevant features are extracted to describe motion, surrounding traffic context, and interaction behavior. These features serve as observations for the DBN.

The thesis studies whether such a probabilistic framework can provide accurate, interpretable, and computationally efficient short-horizon predictions of human driving actions. In particular, it examines the suitability of the dataset, the structure of the DBN, the usefulness of the selected observation features, the quality of future action prediction, the interpretability of the learned states, and the runtime performance of the model for online use.

2. Theoretical Background

2.1. Probability Theory

Human driving behaviour is inherently stochastic. In a traffic scenario, different drivers may interpret the same situation differently depending on their perception, experience, and intentions. As a result, multiple maneuvers may be possible under the same traffic conditions. For example, a driver approaching a slower vehicle may choose to brake, maintain speed, or initiate a lane change. Since such decisions cannot be determined deterministically, behaviour prediction must account for uncertainty. Probability theory provides a mathematical framework for representing and reasoning about uncertainty. Instead of predicting a single deterministic outcome, probabilistic models represent a distribution over possible outcomes and their likelihoods. This allows the model to capture the variability and uncertainty inherent in human decision-making.

In probability theory, each uncertain quantity is represented as a Random Variable (RV) [10]. In the context of driver behaviour modelling, variables such as vehicle velocity, acceleration, or lane position can be treated as observable RVs. In addition, latent quantities such as driver style or driving maneuver intention can also be represented as RVs that must be inferred from observations.

Complex systems require modelling interactions between many RVs. When several RVs are considered together, their behaviour can be described using a joint probability distribution [10]. For two RVs, O and Z, representing the observable RV and latent RV, the joint probability is written as $P(O, Z)$ and represents the probability that both variables take specific values at the same time. This idea can be extended to any number of variables. In driver behaviour modelling, joint probability is important because observable variables and latent variables influence each other and must often be described together within a unified probabilistic framework.

It is also important to describe dependencies between RVs. This is done using conditional probability [10]. The conditional probability of O given Z is defined as

$$P(O \mid Z) = \frac{P(O, Z)}{P(Z)},$$

provided that $P(Z) > 0$. Conditional probability describes how the probability of one RV changes when information about another RV is known. This concept is essential because the probability of a future driving maneuver is estimated conditioned on the observed traffic context during inference. During model learning, the probability of observations conditioned on latent states is also required.

Another important concept is marginal probability, which gives the probability of a subset of RVs irrespective of the values of the others [10]. For example, if the joint distribution $P(O, Z)$ is known, the marginal probability of Z can be obtained by summing over all possible values of O:

$$P(Z) = \sum_{O} P(O, Z).$$

In addition to describing dependencies between RVs, it is also useful to identify situations where variables do not influence each other. Two RVs are said to be independent if the value of one variable does not affect the probability of the other. Formally, two variables A and B are independent if their joint probability can be written as

$$P(A, B) = P(A)P(B).$$

In many practical problems, variables are not completely independent but may become independent when conditioned on another variable. This property is known as conditional independence. Two variables A and B are conditionally independent given a third variable C if

$$P(A, B \mid C) = P(A \mid C)P(B \mid C).$$

Conditional independence is a key concept in probabilistic graphical models because it allows complex joint probability distributions to be represented using simpler conditional relationships between variables [11].

2.1.1. Bayes' Theorem

A fundamental principle in probabilistic reasoning is Bayes' theorem, which allows beliefs about uncertain events to be updated when new observations become available [10]. It relates conditional and marginal probabilities and is expressed as

$$P(Z \mid O) = \frac{P(O \mid Z)P(Z)}{P(O)} \tag{2.1}$$

Here, $P(Z)$ is called the prior probability. The prior represents the initial belief about a variable before observing any new data. In the context of driver behaviour modelling, the prior can represent the initial probability of different latent driver states before considering the current traffic situation. The term $P(O \mid Z)$ is called the likelihood. The likelihood describes how probable the observed data O is, given a particular latent state Z. In driver behaviour prediction, it measures how compatible the observed vehicle motion and traffic features are with a given latent driver state. The term $P(O)$ is called the evidence (or marginal likelihood). It represents the total probability of observing the data O under all possible values of Z. The evidence acts as a normalization factor that ensures the resulting probabilities sum to one. Finally, $P(Z \mid O)$ is called the posterior probability. The posterior represents the updated belief about Z after observing O. In behaviour modelling, the posterior corresponds to the updated probability distribution over the latent driver state given the observed traffic context.

Bayes' theorem therefore provides a systematic way to update probabilistic beliefs when new information becomes available. This principle forms the basis for probabilistic inference in many models used for behaviour prediction, including DBN.

2.2. Probabilistic Graphical Models

Probabilistic graphical models provides an efficient way to represent complex joint probability distributions, whixh may have large number of RVs. This framework uses nodes and edges to represent RVs and dependencies respectively. Conditional independence is then used to factorise the joint distribution. There are 2 types of grapghical models:

1. Directed graphical models, which has edges with direction to show the dependency between 2 RVs.
2. Undirected graphical models which has edges without direction to show the dependency between 2 RVs.

In this thesis we will use Directed graphical model as it allows factorisation

2.3. Bayesian Networks

2.4. Hidden Markov Models

2.5. Dynamic Bayesian Networks

2.6. Inference in DBN

2.7. Parameter Learning with Expectation-Maximization

3. Conclusions

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A. Example appendix chapter

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